

Homework # 4. Due : February 26, 2025

1. Determine the ionization energy of the He^+ ion. Also calculate the minimum frequency a photon must have to cause ionization.

2 In the lithium atom ($Z = 3$), the energy of the outer electron is approximated as

$$E = -\frac{(Z - \sigma)^2 13.6 \text{ eV}}{n^2}$$

where σ is the screening constant. If the measured ionization energy is 5.39 eV, what is the value of screening constant?

3 An electron in the $n = 2$ state of hydrogen remains there on the average of about 10^{-8} s, before making a transition to $n = 1$ state.

- (i) Estimate the uncertainty in the energy of the $n = 2$ state.
- (ii) What fraction of the transition energy is this?
- (iii) What is the wavelength and width of this line in the spectrum of hydrogen atom?

4 If the interelectronic repulsion in helium is ignored, what would be its ground state energy and wave function?